

# Safety data sheet

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BASF Safety data sheet  
Date / Revised: 06.09.2023  
Product: **Citral Extra**

Version: 3.0

(30035068/SDS\_GEN\_TH/EN)

Date of print: 25.02.2026

## 1. Substance/preparation and manufacturer/supplier identification

**Product name:**  
Citral Extra

Use: Chemical, Chemical for detergents, Cosmetic and oral care chemical, flavoring substance

Manufacturer/supplier:  
BASF (Thai) Limited  
23rd Floor, Emporium Tower, 622, Sukhumvit 24 Rd.,  
Klongton, Klongtoey, Bangkok 10110, THAILAND  
Telephone: +66 2624-1999  
Telefax number: +66 2664-9254  
E-mail address: Thailand-SDS-info@basf.com

Emergency information:  
International emergency number:  
Telephone: +49 180 2273-112

## 2. Hazard identification

### Classification according to UN GHS 2009

Classification of the substance and mixture:  
Hazardous to the aquatic environment - acute: Cat.2  
Skin sensitization: Cat.1  
Acute toxicity: Cat.5 (dermal)  
Acute toxicity: Cat.5 (oral)  
Eye irritation: Cat.2A  
Skin irritation: Cat.2

Label elements and precautionary statement:

Pictogram:



Signal Word:  
Warning

## Hazard Statement:

|             |                                                      |
|-------------|------------------------------------------------------|
| H319        | Causes serious eye irritation.                       |
| H315        | Causes skin irritation.                              |
| H317        | May cause an allergic skin reaction.                 |
| H303 + H313 | May be harmful if swallowed or in contact with skin. |
| H401        | Toxic to aquatic life.                               |

## Precautionary Statements (Prevention):

|      |                                                                        |
|------|------------------------------------------------------------------------|
| P280 | Wear protective gloves and eye protection or face protection.          |
| P261 | Avoid breathing mist or vapour or spray.                               |
| P273 | Avoid release to the environment.                                      |
| P272 | Contaminated work clothing should not be allowed out of the workplace. |
| P264 | Wash contaminated body parts thoroughly after handling.                |

## Precautionary Statements (Response):

|                    |                                                                                                                                  |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------|
| P305 + P351 + P338 | IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. |
| P302 + P352        | IF ON SKIN: Wash with plenty of soap and water.                                                                                  |
| P333 + P313        | If skin irritation or rash occurs: Get medical attention.                                                                        |
| P301 + P312        | IF SWALLOWED: Call a POISON CENTER or doctor/physician if you feel unwell.                                                       |
| P302 + P312        | IF ON SKIN: Call a POISON CENTER or a doctor/physician if you feel unwell.                                                       |
| P332 + P313        | If skin irritation occurs: Get medical attention.                                                                                |
| P362 + P364        | Take off contaminated clothing and wash it before reuse.                                                                         |
| P337 + P313        | If eye irritation persists: Get medical attention.                                                                               |

## Precautionary Statements (Disposal):

|      |                                                                                   |
|------|-----------------------------------------------------------------------------------|
| P501 | Dispose of contents and container to hazardous or special waste collection point. |
|------|-----------------------------------------------------------------------------------|

Other hazards which do not result in classification:

When finely distributed on porous material, self-ignition is possible.

### 3. Composition/information on ingredients

#### Chemical nature

Substance nature: Substance

3,7-dimethyl-2,6-octadien-1-ol

CAS Number: 5392-40-5

#### Hazardous ingredients

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**3,7-dimethyl-2,6-octadien-1-al**

Content (W/W):  $\geq 75\%$  -  $\leq 100\%$

CAS Number: 5392-40-5

Acute Tox.: Cat. 5 (oral)

Acute Tox.: Cat. 5 (dermal)

Skin Corr./Irrit.: Cat. 2

Eye Dam./Irrit.: Cat. 2A

Skin Sens.: Cat. 1

Aquatic Acute: Cat. 2

**3-methylbut-2-en-1-ol**

Content (W/W):  $> 0\%$  -  $< 0.2\%$

CAS Number: 556-82-1

Flam. Liq.: Cat. 3

Acute Tox.: Cat. 4 (oral)

Acute Tox.: Cat. 5 (dermal)

Skin Corr./Irrit.: Cat. 1C

Eye Dam./Irrit.: Cat. 1

Aquatic Acute: Cat. 3

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## 4. First-Aid Measures

General advice:

Remove contaminated clothing.

If inhaled:

Keep patient calm, remove to fresh air, seek medical attention.

On skin contact:

Wash thoroughly with soap and water

On contact with eyes:

Wash affected eyes for at least 15 minutes under running water with eyelids held open, consult an eye specialist.

On ingestion:

Immediately rinse mouth and then drink 200-300 ml of water, seek medical attention.

Note to physician:

Symptoms: Information, i.e. additional information on symptoms and effects may be included in the GHS labeling phrases available in Section 2 and in the Toxicological assessments available in Section 11.

Treatment: Treat according to symptoms (decontamination, vital functions), no known specific antidote.

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## 5. Fire-Fighting Measures

Suitable extinguishing media:

dry powder, foam, carbon dioxide, water spray

Unsuitable extinguishing media for safety reasons:

water jet

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**Specific hazards:**

carbon oxides, harmful vapours

The substances/groups of substances mentioned can be released in case of fire.

**Special protective equipment:**

Wear a self-contained breathing apparatus.

**Further information:**

Collect contaminated extinguishing water separately, do not allow to reach sewage or effluent systems. Dispose of fire debris and contaminated extinguishing water in accordance with official regulations. Cool endangered containers with water-spray.

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## 6. Accidental Release Measures

**Personal precautions:**

Use personal protective clothing. Information regarding personal protective measures, see section 8. Ensure adequate ventilation. Do not breathe vapour/spray. Avoid contact with the skin, eyes and clothing.

**Environmental precautions:**

Do not discharge into drains/surface waters/groundwater. Inform authorities in the event of product spillage to water courses or sewage systems.

**Methods for cleaning up or taking up:**

For small amounts: Pick up with suitable absorbent material. Do not use saw-dust or other combustible substances as an absorbant during cleanup.

For large amounts: Dike spillage. Pump off product.

Dispose of absorbed material in accordance with regulations. Mop up spills with non-flammable adsorbents (e.g. vermiculite, spill mats). Soiled textiles / cleaning rags / adsorbents and Silica are capable of self ignition and should be wetted with water and must be disposed of in a safe manner.

**Additional information:** When finely distributed on porose material, self-ignition is possible. Soiled textiles/cleaning rags made of natural fibres (e.g. of pure wool or of pure cotton) are capable of ignition and should not be used and/or must be desposed of in a safe manner.

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## 7. Handling and Storage

**Handling**

Ensure thorough ventilation of stores and work areas. Wear suitable protective clothing and eye/face protection. Avoid contact with the skin, eyes and clothing. Keep container tightly sealed. This product may cause irritations; wash your hands after every contact.

**Protection against fire and explosion:**

Risk of self-ignition when a large surface area is produced due to fine dispersion. Soiled textiles / cleaning rags / adsorbents and Silica are capable of self ignition and should be wetted with water and must be disposed of in a safe manner. Avoid all sources of ignition: heat, sparks, open flame. Take precautionary measures against static discharges.

### Storage

Further information on storage conditions: Keep container tightly closed in a cool, well-ventilated place. Protect from the effects of light.

## 8. Exposure controls and personal protection

### Components with occupational exposure limits

3,7-dimethyl-2,6-octadien-1-al, 5392-40-5;

TWA value 5 ppm (ACGIHTLV), Inhalable fraction and vapor

Skin Designation (ACGIHTLV), Inhalable fraction and vapor

Danger of cutaneous absorption

Skin Designation (ACGIHTLV), Inhalable fraction and vapor

Danger of cutaneous absorption

### Personal protective equipment

Respiratory protection:

Wear respiratory protection if ventilation is inadequate. Gas filter for gases/vapours of organic compounds (boiling point >65 °C, e. g. EN 14387 Type A)

Hand protection:

Suitable materials also with prolonged, direct contact (Recommended: Protective index 6, corresponding > 480 minutes of permeation time according to EN ISO 374-1):

butyl rubber (butyl) - 0.7 mm coating thickness

fluoroelastomer (FKM) - 0.7 mm coating thickness

Supplementary note: The specifications are based on tests, literature data and information of glove manufacturers or are derived from similar substances by analogy. Due to many conditions (e.g. temperature) it must be considered, that the practical usage of a chemical-protective glove in practice may be much shorter than the permeation time determined through testing.

Manufacturer's directions for use should be observed because of great diversity of types.

Eye protection:

Safety glasses with side-shields (frame goggles) (e.g. EN 166)

Body protection:

Body protection must be chosen depending on activity and possible exposure, e.g. apron, protecting boots, chemical-protection suit (according to EN 14605 in case of splashes or EN ISO 13982 in case of dust).

General safety and hygiene measures:

Handle in accordance with good industrial hygiene and safety practice. Wearing of closed work clothing is required additionally to the stated personal protection equipment. Avoid contact with the skin, eyes and clothing. Do not breathe vapour/spray. No eating, drinking, smoking or tobacco use at the place of work. Hands and/or face should be washed before breaks and at the end of the shift. Store work clothing separately.

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## 9. Physical and Chemical Properties

|                               |                                                                                                                                        |                                                               |
|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|
| Form:                         | liquid                                                                                                                                 |                                                               |
| Colour:                       | colourless to yellowish                                                                                                                |                                                               |
| Odour:                        | of lemon                                                                                                                               |                                                               |
| Odour threshold:              | < 100 ppm                                                                                                                              |                                                               |
| pH value:                     | not applicable                                                                                                                         |                                                               |
| glass transition temperature: | -115 °C                                                                                                                                | (DSC (DIN 51007))                                             |
| Boiling point:                | approx. 230 °C<br>(1,013 hPa)<br>The substance / product<br>decomposes.                                                                | (other)                                                       |
| Flash point:                  | 98 °C<br>Literature data.                                                                                                              | (other)                                                       |
| Evaporation rate:             | Value can be approximated from<br>Henry's Law Constant or vapor<br>pressure.                                                           |                                                               |
| Flammability (solid/gas):     | hardly combustible                                                                                                                     | (derived from flash point)                                    |
| Lower explosion limit:        | For liquids not relevant for<br>classification and labelling., The<br>lower explosion point may be 5 - 15<br>°C below the flash point. |                                                               |
| Upper explosion limit:        | For liquids not relevant for<br>classification and labelling.                                                                          |                                                               |
| Ignition temperature:         | 225 °C<br>Literature data.                                                                                                             | (DIN 51794)                                                   |
| Thermal decomposition:        | approx. 180 °C                                                                                                                         | (DSC (DIN 51007))                                             |
| Self ignition:                | Based on its structural properties the<br>product is not classified as self-<br>igniting.                                              | Test type: Spontaneous self-<br>ignition at room-temperature. |
| SADT:                         | > 75 °C<br>Heat accumulation / Dewar 500 ml (SADT, UN-Test H.4, 28.4.4)                                                                |                                                               |
| Explosion hazard:             | Based on the chemical structure<br>there is no indication of explosive<br>properties.                                                  |                                                               |
| Fire promoting properties:    | Based on its structural properties<br>the product is not classified as<br>oxidizing.                                                   |                                                               |
| Vapour pressure:              | 0.046 hPa<br>(20 °C)                                                                                                                   | (calculated)                                                  |
|                               | 0.071 hPa<br>(25 °C)                                                                                                                   | (calculated)                                                  |

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|                                                     |                                                                      |                                         |
|-----------------------------------------------------|----------------------------------------------------------------------|-----------------------------------------|
|                                                     | 1.003 hPa<br>(59.29 °C)                                              | (measured)                              |
| Density:                                            | 0.89 g/cm <sup>3</sup><br>(20 °C)                                    |                                         |
| Relative density:                                   | Literature data.<br>0.89<br>(20 °C)                                  | (other)                                 |
| Relative vapour density (air):                      | Literature data.<br>5.24<br>(20 °C)<br>Heavier than air.             | (calculated)                            |
| Solubility in water:                                | moderately soluble<br>0.42 g/l<br>(25 °C)                            |                                         |
| Partitioning coefficient n-octanol/water (log Pow): | 2.76<br>(25 °C)                                                      | (OECD Guideline 107)                    |
| Adsorption/water - soil:                            | log KOC: 2.1                                                         | (calculated)                            |
| Surface tension:                                    | Based on chemical structure, surface activity is not to be expected. |                                         |
| Viscosity, dynamic:                                 | 2.15 mPa.s<br>(20 °C)                                                | (calculated (from kinematic viscosity)) |
|                                                     | 1.46 mPa.s<br>(40 °C)                                                | (calculated (from kinematic viscosity)) |
| Viscosity, kinematic:                               | 2.42 mm <sup>2</sup> /s<br>(20 °C)                                   | (OECD 114)                              |
|                                                     | 1.67 mm <sup>2</sup> /s<br>(40 °C)                                   | (OECD 114)                              |

## 10. Stability and Reactivity

Conditions to avoid:

Avoid all sources of ignition: heat, sparks, open flame. See SDS section 7 - Handling and storage.

Thermal decomposition: approx. 180 °C (DSC (DIN 51007))

Substances to avoid:

acids, bases

Corrosion to metals: No corrosive effect on metal.

Hazardous reactions:

Self-ignition is possible when finely distributed on flammable surfaces in the presence of air.

Hazardous decomposition products:

No hazardous decomposition products if stored and handled as prescribed/indicated.

Chemical stability:

The product is stable if stored and handled as prescribed/indicated.

Reactivity:

No hazardous reactions if stored and handled as prescribed/indicated.

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## 11. Toxicological Information

### Routes of exposure

#### Acute oral toxicity

Experimental/calculated data:

LD50rat (oral): approx. 6,800 mg/kg (BASF-Test)

#### Acute dermal toxicity

LD50 rat (dermal): > 2,000 mg/kg (BASF-Test)

#### Assessment of acute toxicity

Of low toxicity after single ingestion. Of low toxicity after short-term skin contact.

#### Symptoms

Information, i.e. additional information on symptoms and effects may be included in the GHS labeling phrases available in Section 2 and in the Toxicological assessments available in Section 11.

#### Irritation

Assessment of irritating effects:

Skin contact causes irritation. Eye contact causes irritation.

Experimental/calculated data:

Skin corrosion/irritation rabbit: Irritant. (BASF-Test)

Serious eye damage/irritation rabbit: Irritant. (BASF-Test)

#### Respiratory/Skin sensitization

Assessment of sensitization:

Caused skin sensitization in animal studies. Caused sensitization in humans.

Experimental/calculated data:

Guinea pig maximization test guinea pig: skin sensitizing

#### Germ cell mutagenicity

Assessment of mutagenicity:

The substance was not mutagenic in bacteria. In the majority of tests performed (mammalian cell culture) a mutagenic effect was not found. A mutagenic effect was also not observed in in-vivo assays.

#### Carcinogenicity

Assessment of carcinogenicity:

Results from a number of long-term carcinogenity studies are available. Taking into account all of the information, there is no indication that the substance itself is carcinogenic.

### **Reproductive toxicity**

Assessment of reproduction toxicity:

The results of animal studies gave no indication of a fertility impairing effect.

### **Developmental toxicity**

Assessment of teratogenicity:

Animal studies gave no indication of a developmental toxic effect at doses that were not toxic to the parental animals.

### **Specific target organ toxicity (single exposure)**

Based on available data, the classification criteria are not met.

### **Repeated dose toxicity and Specific target organ toxicity (repeated exposure)**

Assessment of repeated dose toxicity:

Prolonged repeated exposure caused inflammable degenerative processes in the respiratory tract of rats. Causes irritating effects at esophagus and the gastro-intestinal tract.

### **Aspiration hazard**

No aspiration hazard expected.

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## **12. Ecological Information**

### **Ecotoxicity**

Assessment of aquatic toxicity:

Acutely toxic for aquatic organisms. Depending on local conditions and existing concentrations, disturbances in the biodegradation process of activated sludge are possible.

Toxicity to fish:

LC50 (96 h) 6.8 mg/l, *Leuciscus idus* (DIN 38412 Part 15, static)

The product has low solubility in the test medium. An aqueous solution prepared with solubilizers has been tested. The details of the toxic effect relate to the nominal concentration.

Aquatic invertebrates:

EC50 (48 h) approx. 7 mg/l, *Daphnia magna* (Directive 79/831/EEC, static)

The product has low solubility in the test medium. An aqueous solution prepared with solubilizers has been tested. The details of the toxic effect relate to the nominal concentration.

Aquatic plants:

EC50 (72 h) 103.8 mg/l (growth rate), *Scenedesmus subspicatus* (DIN 38412 Part 9, static)

The product has low solubility in the test medium. An aqueous solution prepared with solubilizers has been tested. The details of the toxic effect relate to the nominal concentration.

Microorganisms/Effect on activated sludge:

EC50 (30 min) 2,100 mg/l, *Pseudomonas putida* (DIN 38412 Part 27 (draft), aquatic)

The product has low solubility in the test medium. An aqueous solution prepared with solubilizers has been tested. The details of the toxic effect relate to the nominal concentration.

EC20 (30 min) approx. 68 mg/l, activated sludge, domestic (OECD Guideline 209, aquatic)

Assessment of terrestrial toxicity:  
Study scientifically not justified.

### **Mobility**

Assessment transport between environmental compartments:  
The substance will slowly evaporate into the atmosphere from the water surface.  
Adsorption to solid soil phase is not expected.

### **Persistence and degradability**

Elimination information:

92 % BOD of the ThOD (28 d) (OECD 301C; ISO 9408; 92/69/EEC, C.4-F) (aerobic, activated sludge, domestic)

> 90 % BOD of the ThOD (28 d) (OECD 301F; ISO 9408; 92/69/EEC, C.4-D) (aerobic, activated sludge, domestic)

Assessment of stability in water:

Substance is readily biodegradable, therefore hydrolysis is not expected to be relevant.  
Study scientifically not justified.

### **Bioaccumulation potential**

Assessment bioaccumulation potential:

Because of the n-octanol/water distribution coefficient (log Pow) accumulation in organisms is not to be expected.

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## **13. Disposal Considerations**

Observe national and local legal requirements.

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## **14. Transport Information**

### **Domestic transport:**

|                              |                                                                |
|------------------------------|----------------------------------------------------------------|
| UN number or ID number       | Not classified as a dangerous good under transport regulations |
| UN proper shipping name:     | Not applicable                                                 |
| Transport hazard class(es):  | Not applicable                                                 |
| Packing group:               | Not applicable                                                 |
| Environmental hazards:       | Not applicable                                                 |
| Special precautions for user | None known                                                     |

### **Sea transport**

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**IMDG**

|                              |                                                                |
|------------------------------|----------------------------------------------------------------|
|                              | Not classified as a dangerous good under transport regulations |
| UN number or ID number:      | Not applicable                                                 |
| UN proper shipping name:     | Not applicable                                                 |
| Transport hazard class(es):  | Not applicable                                                 |
| Packing group:               | Not applicable                                                 |
| Environmental hazards:       | Not applicable                                                 |
|                              | Marine pollutant: no                                           |
| Special precautions for user | None known                                                     |

**Air transport**

IATA/ICAO

|                              |                                                                |
|------------------------------|----------------------------------------------------------------|
|                              | Not classified as a dangerous good under transport regulations |
| UN number or ID number       | Not applicable                                                 |
| Proper shipping name:        | Not applicable                                                 |
| Transport hazard class(es):  | Not applicable                                                 |
| Packing group:               | Not applicable                                                 |
| Environmental hazards:       | Not applicable                                                 |
| Special precautions for user | None known                                                     |

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**15. Regulatory Information****Other regulations**

If other regulatory information applies that is not already provided elsewhere in this safety data sheet, then it is described in this subsection.

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**16. Other Information**

Any other intended applications should be discussed with the manufacturer. Corresponding occupational protection measurements must be followed.

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Vertical lines in the left hand margin indicate an amendment from the previous version.

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